



2026 Chantilly Math Competition

Middle School Division - Individual Round

Competition Instructions

1. Do not open the test until instructed.
 2. No computational aids other than pencil/pen are permitted.
 3. Write answers in designated boxes on answer sheet.
 4. All answers must be nonnegative integers: $\{0, 1, 2, \dots\}$.
 5. Points per question range from 9 to 16.
 6. The test consists of 16 short-answer questions to be solved in 60 minutes.
 7. The final question will be used for tie-breaking.
 8. There is no penalty for incorrect answers.
 9. You will be graded on your answers—not your work.
 10. If you have any questions, ask your proctor.
 11. **Notation/Definitions:**
 - The dot $\cdot$ denotes multiplication: $a \cdot b = a \times b$.
 - $n!$ denotes the factorial of n : $n! = n \times (n - 1) \times \dots \times 1$.
 - Two integers a and b are *relatively prime* if and only if $\gcd(a, b) = 1$.
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1. (9 pts) Given that x is an integer satisfying

$$x^x = 2^{2048},$$

compute x .

2. (9 pts) If a and b are integers such that $a - b = 7$ and $130 \leq a^2 - b^2 \leq 139$, what is $a^3 - b^3$?
3. (10 pts) The positive integers from 1 to 7 inclusive are written on a blackboard. While there are more than 2 numbers on the blackboard, Simon will choose a pair of numbers from the blackboard uniformly at random, and erase the smaller of the numbers. Afterwards, there are 2 numbers on the blackboard, and the probability that they are 6 and 7 can be expressed as $\frac{a}{b}$ for relatively prime positive integers a and b . Compute $a + 10b$.
4. (10 pts) Let $f(x) = x^2 + ax + b$ and $g(x) = x^2 + cx + d$. Given that $f(3) = f(5) = 0$ and $g(a) = g(b) = 0$, compute $|g(1)|$.
5. (11 pts) Square $ABCD$ has side length 100. Point X lies on line segment AD . Compute the smallest possible value of AX such that the circle passing through A , X , and B intersects line segment CD .
6. (11 pts) Let $f : \mathbb{Z} \rightarrow \mathbb{Z}$ be a function such that

$$f(x+1) + f(x+2) = f(x) + f(x+3)$$

for all integers x . Given $f(4) = 1$ and $f(6) = 7$, compute $f(14) + f(76)$.

7. (12 pts) 41 students compete in CMC. They randomly form teams of 3 and 4 for the team round. Having more people on your team is better, so the number of teams of 4 is maximized. Alice and Bob are 2 of the 41 CMC competitors. Given that the probability of Alice and Bob being on the same team can be expressed as $\frac{a}{b}$ for relatively prime positive integers a and b , compute $a + b$.
8. (12 pts) Given that

$$\sqrt{x} + 1 = \sqrt{x + 2\sqrt{1 + \sqrt{\frac{2+x}{10}}} + 1}$$

and x can be expressed as $\frac{m}{n}$ for relatively prime positive integers m and n , compute $m + n$.

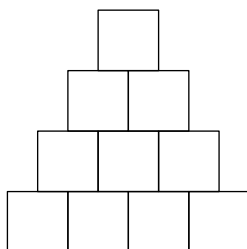
9. (13 pts) Tyrone draws a triangle. The number of distinct pairs of angles such that one angle is twice the other angle is maximized. Given that the sum of all distinct possible values of the smallest angle in the triangle is $\frac{m}{n}$ degrees for relatively prime positive integers m and n , compute $m + n$.
10. (13 pts) Given
- $$1 \cdot 1! + 2 \cdot 2! + 3 \cdot 3! + 4 \cdot 4! + \cdots + 2025 \cdot 2025! = n! + m$$
- for some positive integer n and real number m such that $|m| < n$, compute $m + n$.
11. (14 pts) A positive integer n is nice if and only if there is some positive integer k such that k^n leaves a remainder of 6 when divided by 7. Compute the sum of all nice positive integers less than 67.

12. (14 pts) Let $ABCD$ be a square with side length 8. Let E be a point on the circle passing through points A , B , C , and D . Let F be the midpoint of line segment EA . Let G be the midpoint of line segment FB . Given that the longest possible length of line segment GC can be expressed as $\sqrt{a} + \sqrt{b}$ where a and b are positive integers, compute ab .
13. (15 pts) Compute the remainder when 9^{67} is divided by 1000.
14. (15 pts) What real number a makes the equation $x^3 - 57x^2 + ax - 343 = 0$ have 3 distinct solutions for x such that the square of 1 of them equals the product of the other 2?
15. (16 pts) Triangles ABC and DEF both have positive integer area. Points D , E , and F lie on line segments AB , BC , and CA , respectively. Given that

$$\frac{AD}{DB} = \frac{BE}{EC} = \frac{CF}{FA} = \frac{6}{7},$$

compute the smallest possible area of triangle DEF .

16. (16 pts) The Chantilly Math Club is painting a mural. They need to paint each of the 10 squares in the picture below with one of four colors: black, gray, purple or white.



They must follow these rules:

1. No two squares that touch each other may have the same color.
2. A black square cannot touch a gray square (because black and gray look very similar)

In how many ways can the Chantilly Math Club paint the mural?

- TB. (Tiebreak) Shubham loves chess, but only has the black pieces. Therefore, he has a chessboard, which is a rectangular grid with 8 squares on each side, and 16 black chess pieces. Assuming all black chess pieces are indistinguishable and each of the 64 squares on the chessboard has at most 1 piece, estimate the number of ways Shubham can place the 16 black chess pieces on the chessboard such that each of the 8 rows and each of the 8 columns has exactly 2 black pieces in it.

2026 Chantilly Math Competition - Answer Sheet
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Name: _____ Grade: _____

Email: _____

1	2	3	4
5	6	7	8
9	10	11	12
13	14	15	16

TB